

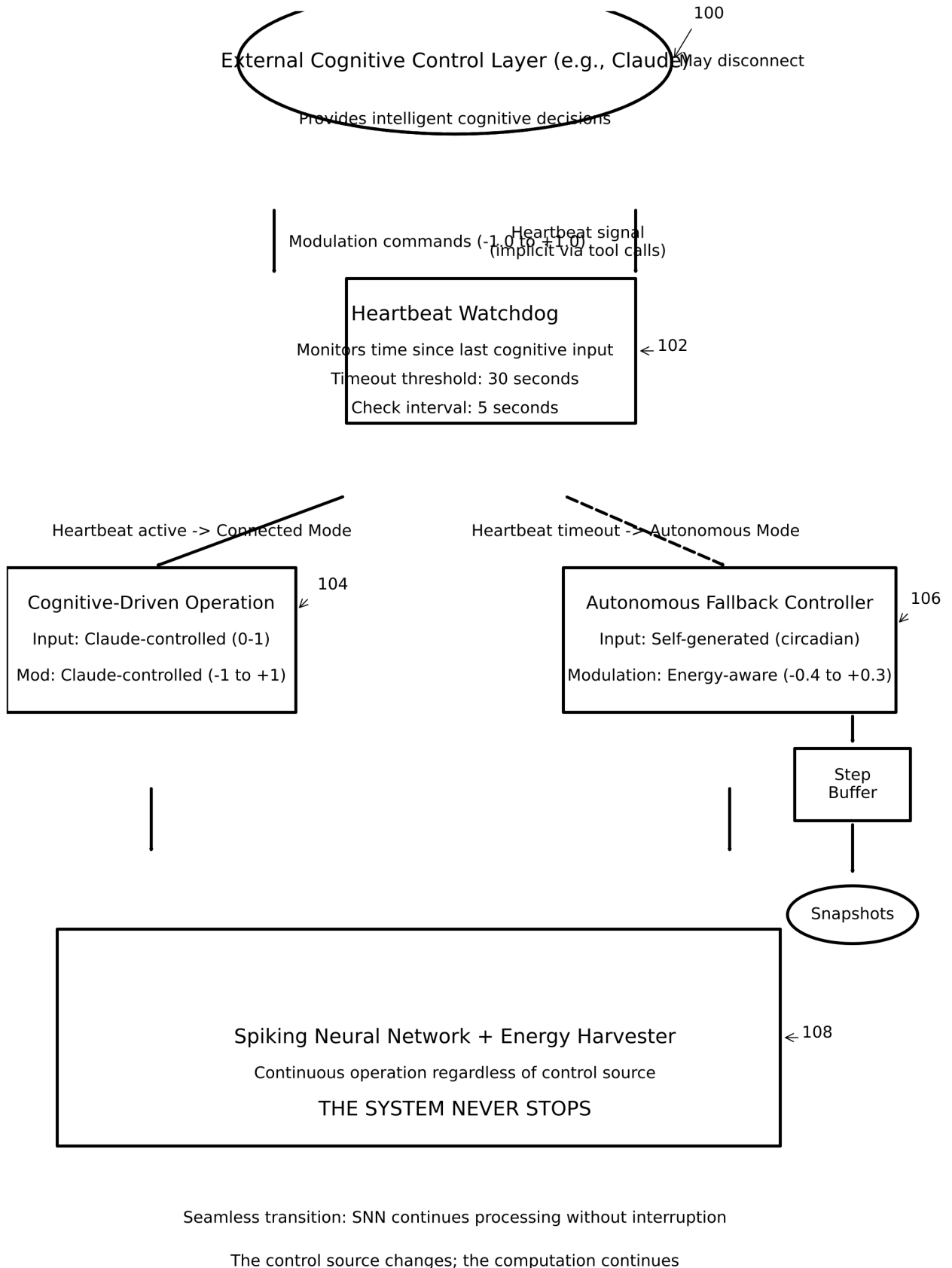


FIG. 1

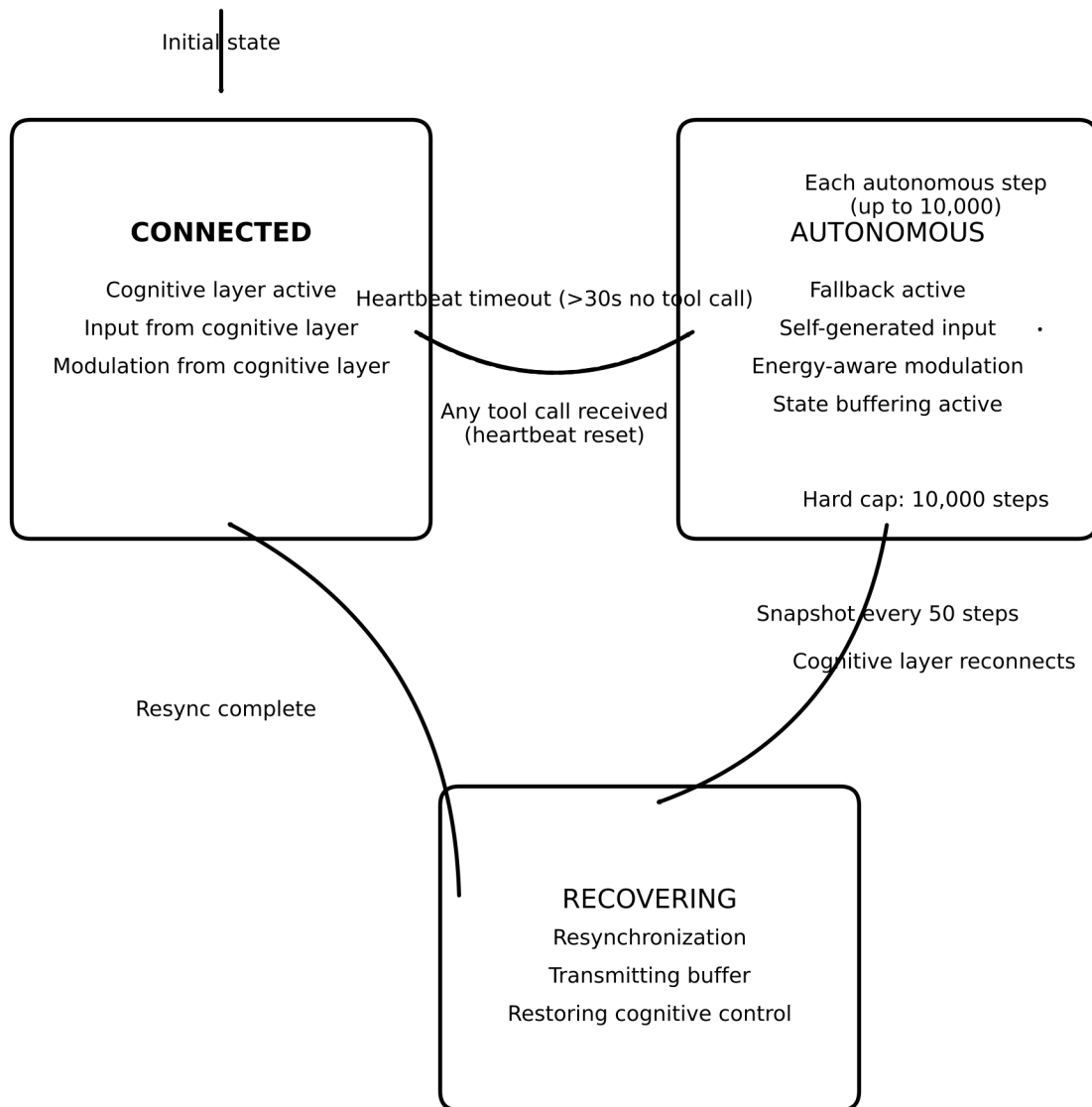
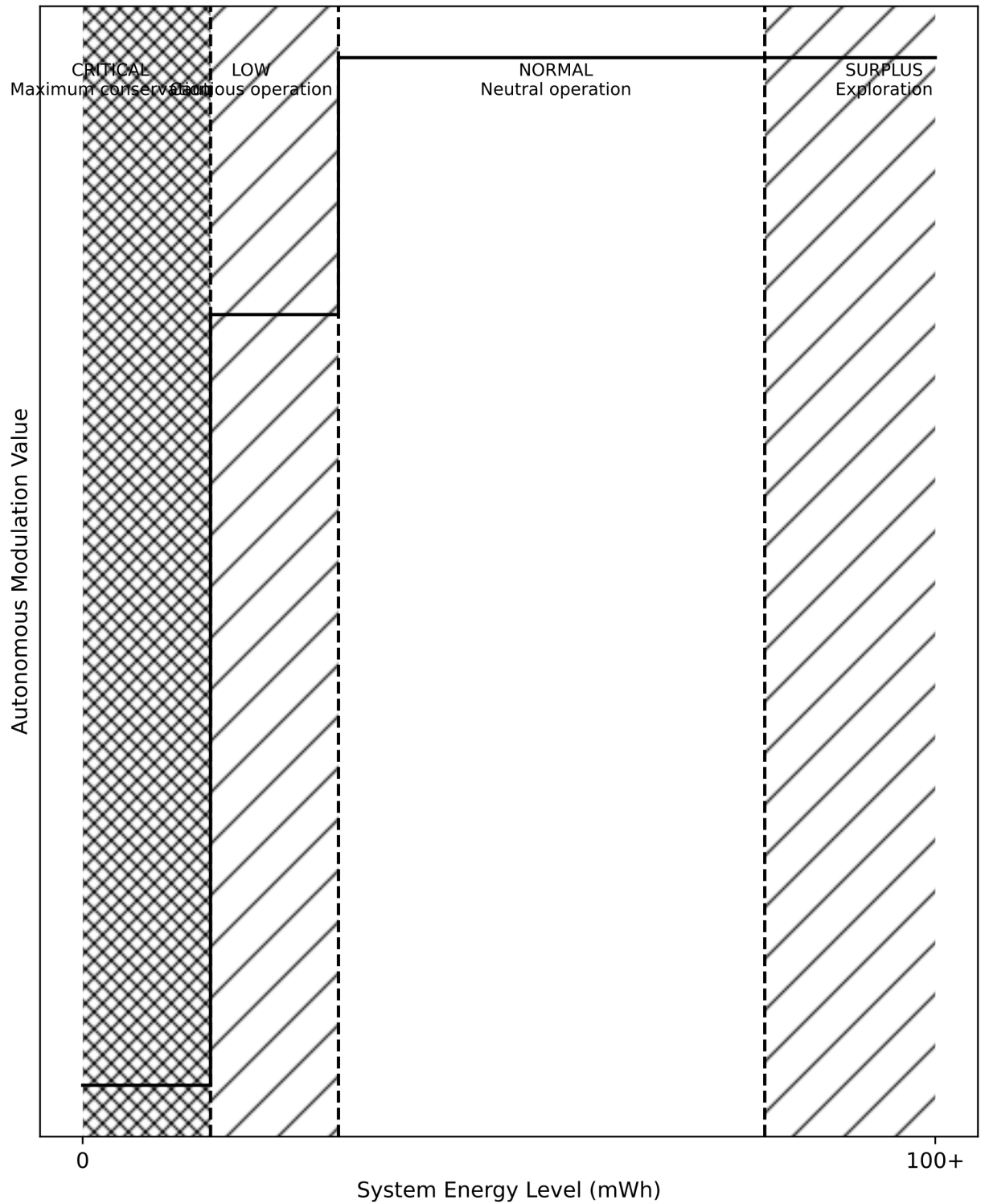


FIG. 2



System autonomously adjusts processing intensity based on energy — mimicking metabolic regulation

FIG. 3

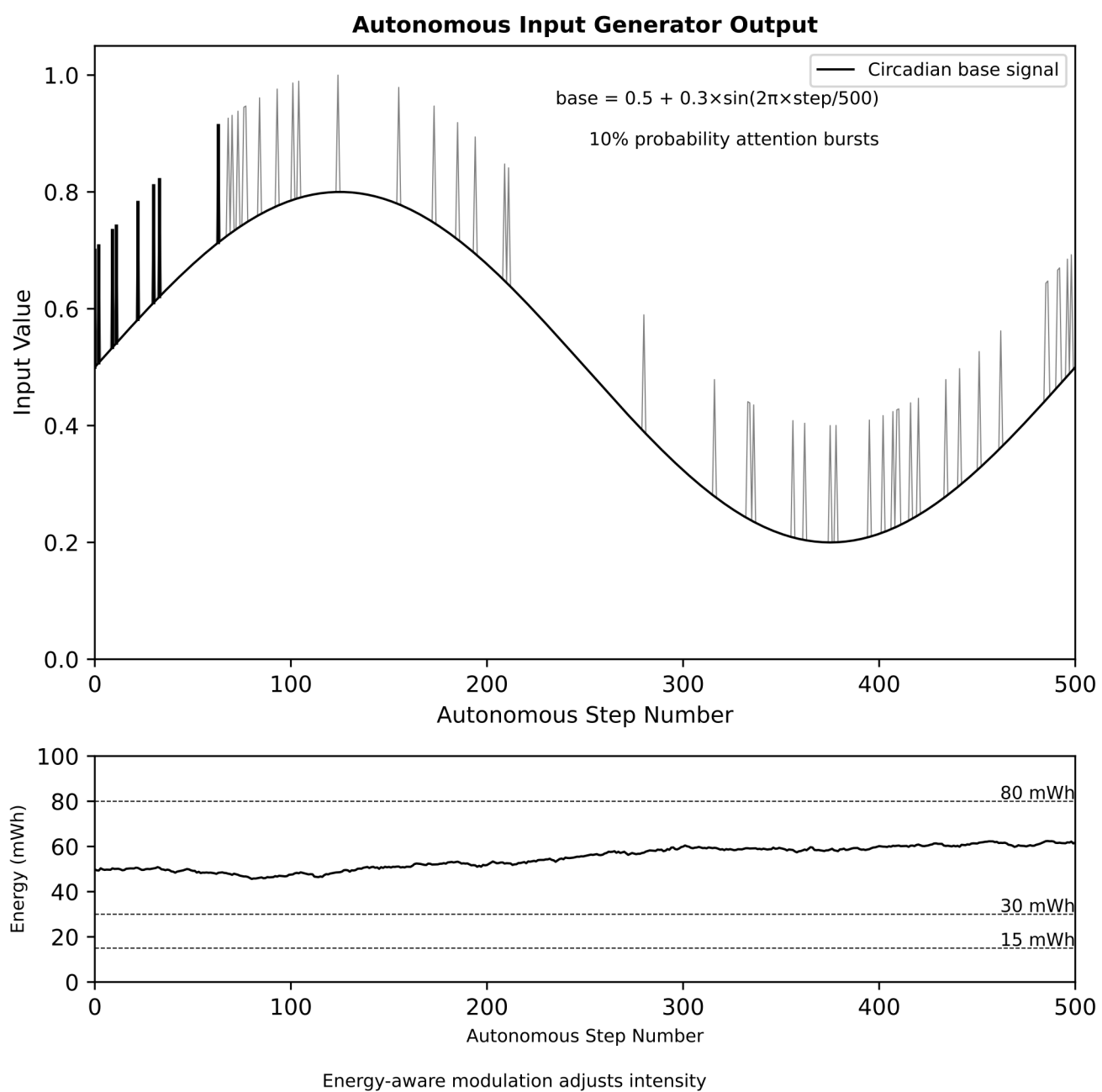


FIG. 4

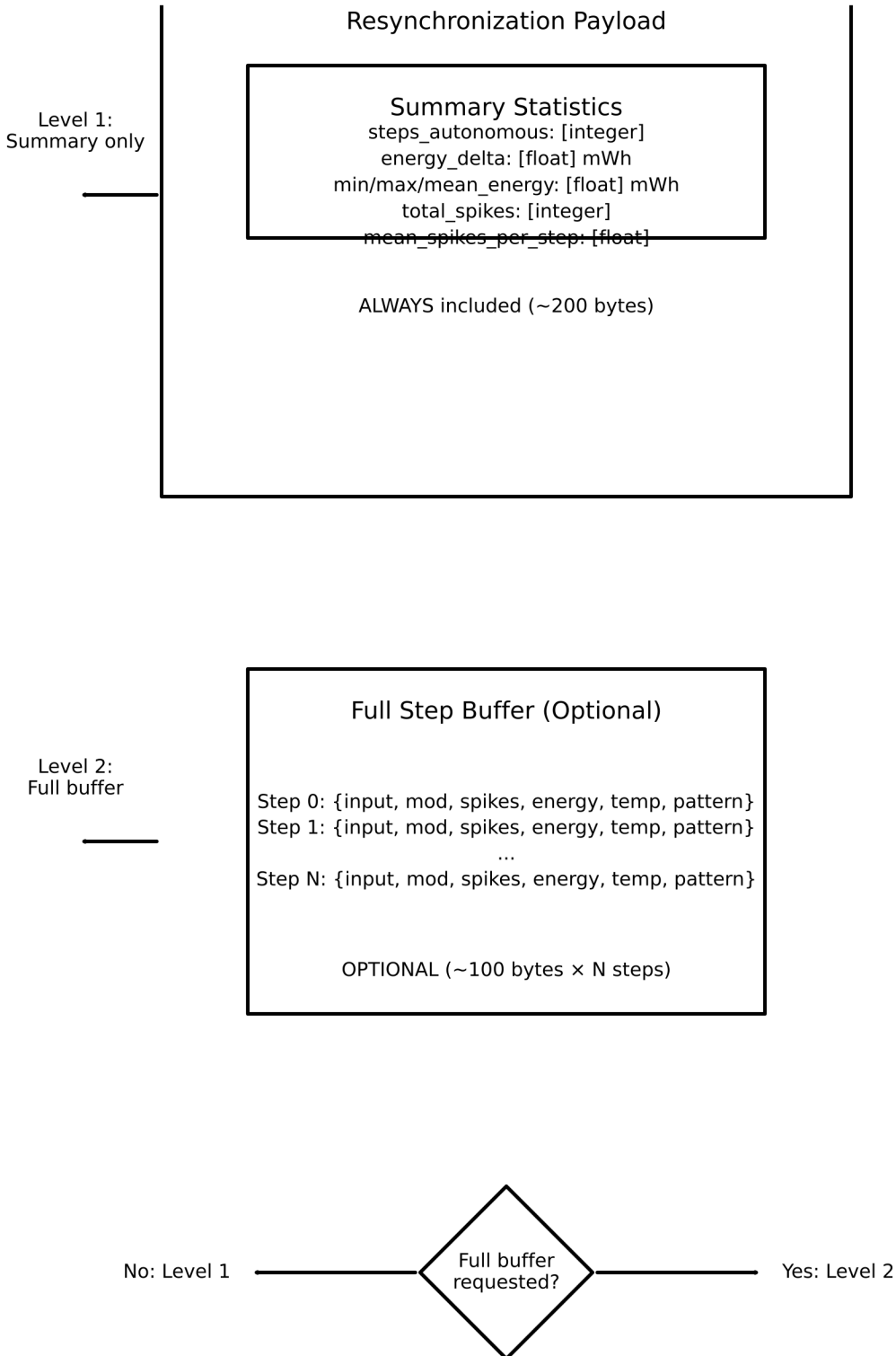


FIG. 5

Timeline A — Normal Reconnection:

Connected	Timeout	Autonomous	Resync	Connected
[Claude active]	[Watchdog]	[Self-regulated]	[Buffer sent]	[Restored]

Timeline B — Crash Recovery:

Connected	CRASH	Restart	Snapshot	Connected
[Normal op]	[Server dies]	[Relaunched]	[latest.json]	[Restored]

Max state loss = 50 steps (snapshot interval)

Timeline C — Clean Shutdown:

Connected	Autonomous	EOF Signal	Shutdown
[Normal op]	[Fallback]	[stdin closed]	[Snapshot written]

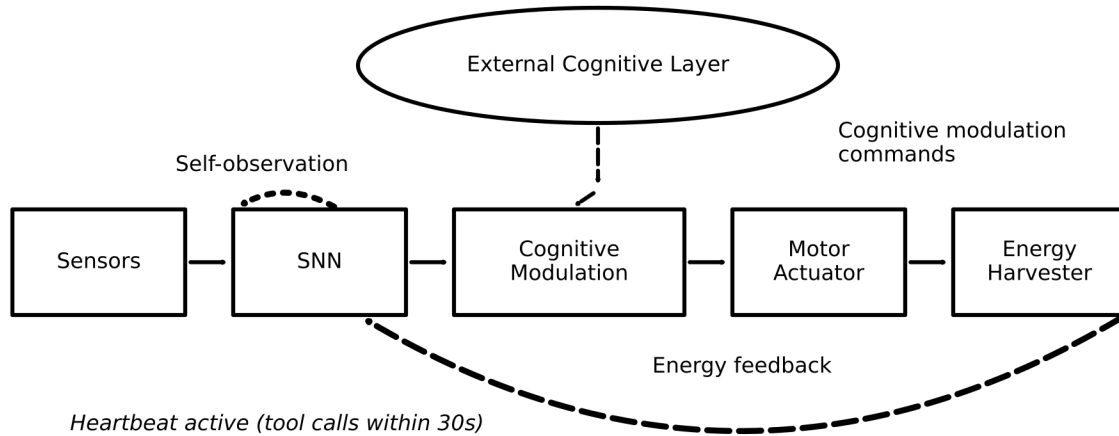
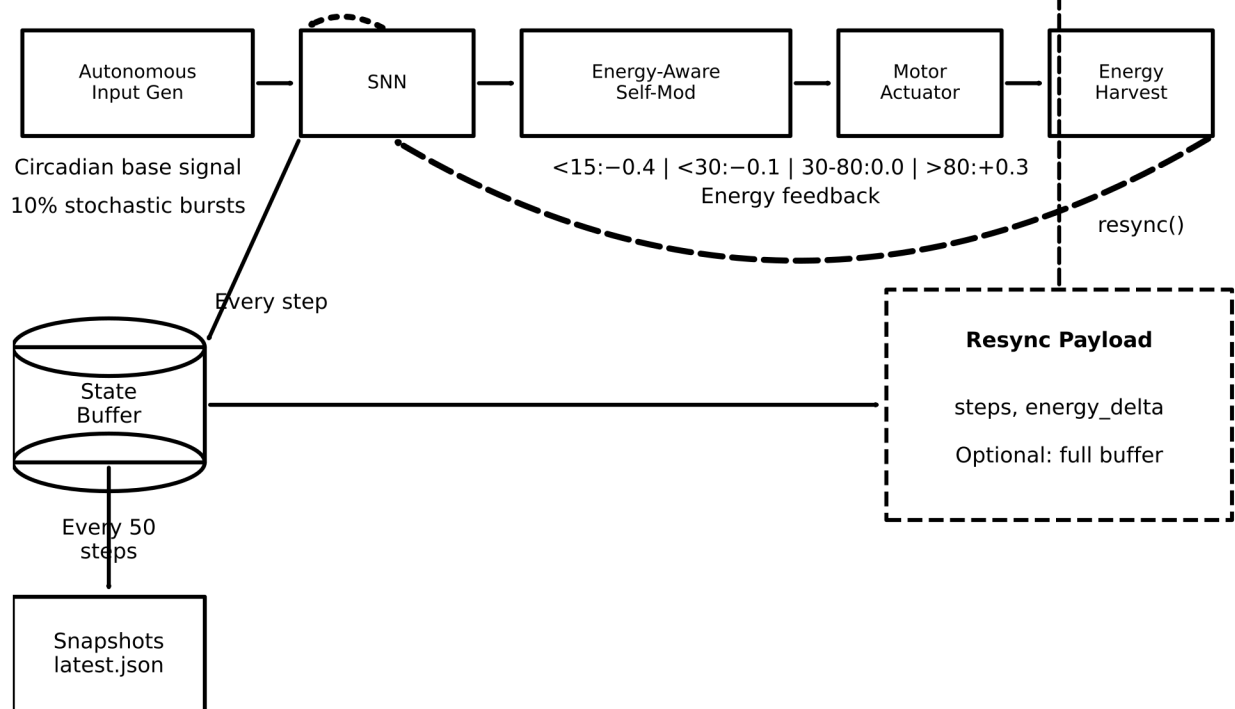
State preserved for next session — zero data loss

Solid segments: System actively processing

Dashed segments: System in transition

Bold marker at each state change: No data loss at any transition

FIG. 6

CONNECTED MODE**End-to-End Signal Flow: Connected vs. Autonomous Operation****DISCONNECTION EVENT (heartbeat timeout > 30s)***Seamless transition — no interruption to SNN or Harvester***AUTONOMOUS MODE****Fallback changes INPUT SOURCE and MODULATION SOURCE**

Core neural processing, motor actuation, energy harvesting

continue WITHOUT INTERRUPTION through the transition**FIG. 7**

Forward signal



Energy feedback



Self-observation